

Numerical Method

- ① Cramer's Rule:
- Find determinant of the matrix x
 - Find variable specific determinant {calculator} Like (Dx_1, Dx_2, Dx_3)
 - $x_1 = \frac{Dx_1}{D}$, $x_2 = \frac{Dx_2}{D}$, $x_3 = \frac{Dx_3}{D}$

- ② Iteration method:
- given $f(x) = 0$
 - convert $x = g(x)$
 - $x_{n+1} = g(x_n)$
 - until $|x_{n+1} - x_n| < \epsilon$
- if $|g'(x)| < 1$ convergence
else divergence

- ③ Bisection:
- root exist if $f(a) \cdot f(b) < 0$ [guess a, b]
 - compute root approximation midpoint
 - check for convergence $\rightarrow$ if $|f(x_m)| < \text{tolerance}$

$$x_m = \frac{x_l + x_r}{2}$$

- if $f(x_l) \cdot f(x_m) < 0$ set $x_r = x_m$
- if $f(x_l) \cdot f(x_m) > 0$ set $x_l = x_m$
- if $f(x_l) \cdot f(x_m) = 0$; x_m is the root

- ④ False position: similar to bisection - just find $x_m \rightarrow$

$$x_m = b - \frac{f(b)(a-b)}{f(a) - f(b)}$$

⑤ Newton Raphson method:

- given $f(x)$ or make $x^2 - A = 0$ ^{$A=2$} [find $\sqrt{2}$]
- find $f'(x)$ [first derivative]
- x_n = current value [need initial guess]
- $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
- until convergence

⑥ Gauss elimination:

{calculator}

- step:
- Convert into upper triangular matrix
 - Back substitution

⑦ Gauss jordan :

{calculator}

- Convert into identity matrix

⑧ Interpolation:

difference table:

reference row	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
	1	2	2	1	0	1	2			
	2	4	3	1		2	4	2		
	3	7	4			3	7	3	1	
	4	11		diagonal		4	11	4	1	0 (R.R.)

horizontal difference ←

diagonal

Forward formula:

$$f(x) \approx y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots$$

$$u = \frac{x - x_0}{h}$$

$\rightarrow$ reference row x value
 $\rightarrow$ x values gap

Backward formula:

$$f(x) \approx y_n + u \nabla y_n + \frac{u(u+1)}{2!} \nabla^2 y_n + \frac{u(u+1)(u+2)}{3!} \nabla^3 y_n + \dots$$

$$u = \frac{x - x_n}{h}$$

⑨ Trapezoidal rule: $I = \int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$

⑩ Simpson's 1/3 : $I = \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)]$

⑪ Simpson's 3/8 : $I = \frac{b-a}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$

⑫ Euler's method: • given $\frac{dy}{dx} = f(x, y)$; $y(x_0) = y_0$

$$y_{n+1} = y_n + f(x_n, y_n) \cdot h$$

⑬ Runge - kutta method: • $k_1 = f(x_n, y_n)$

$$• k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1)$$

$$• k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_2)$$

$$• k_4 = f(x_n + h, y_n + h k_3)$$

$$• y_{n+1} = y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

⑭ Least square regression:

$$\bullet y = \beta_0 + \beta_1 x + e$$

$$\bullet e = y - \beta_0 - \beta_1 x$$

$$\bullet \sum |e_i|^2 = \sum (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\beta_0 = \bar{y} - b\bar{x}$$

$$\beta_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$\beta_0 = \bar{y} - b\bar{x}$$

$$\bullet \text{standard deviation, } S_y = \sqrt{\frac{st}{n-1}} \quad \left| \quad st = \sum (y_i - \bar{y})^2 \right.$$

$$\bullet \text{standard error, } S_{y/x} = \sqrt{\frac{sr}{n-(m+1)}} \quad \left| \quad sr = \sum (y - \beta_0 - \beta_1 x)^2 \right.$$

$$\bullet \text{coefficient of deviation, } r^2 = \frac{st - sr}{st}$$

$$\bullet \text{correlation coefficient, } r$$

⑮ Polynomial: same way - find $\beta_0, \beta_1, \beta_2$

Logistic regression

$$(16) \quad p = \frac{1}{1 + e^{-z}} \quad \left| \quad z = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots$$

$$(17) \quad \text{stochastic gradient decent:} \quad \left| \begin{array}{l} \bullet L(w) = (y - wx)^2 \\ \bullet \frac{dL}{dw} = -2x(y - wx) \\ \bullet w = w - \eta \frac{\partial L(w)}{\partial w} \end{array} \right.$$

(18) Lagrange's interpolation:

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 \\ + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3 + \dots$$

(19) Picard's method:

$$\begin{aligned} \bullet y_1 &= y_0 + \int_{x_0}^x f(x, y_0) dx \quad [\text{first approximation}] \\ \bullet y_2 &= y_0 + \int_{x_0}^x f(x_1, y_1) dx \quad [\text{2nd approximation}] \\ \bullet y_3 &= y_0 + \int_{x_0}^x f(x_2, y_2) dx \quad [\text{3rd}] \end{aligned}$$

• $\frac{dy}{dx} = 1 + xy$ and $y(0) = 1$. calculate $y(0.1)$, $y(0.2)$

① $f(x, y) = 1 + xy$ ② $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$ ③ $y(0.1) = ?$
 $y(0.2) = ?$

First:

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$= 1 + \int_0^x 1 + xy_0 dx$$

$$= 1 + \int_0^x 1 + x(1) dx$$

$$= 1 + x + \frac{x^2}{2}$$

second:

$$y_2 = y_0 + \int_{x_0}^x f(x, y_1) dx$$

$$= 1 + \int_0^x 1 + xy_1 dx$$

$$= 1 + \int_0^x 1 + x\left(1 + x + \frac{x^2}{2}\right) dx$$

$$= 1 + \int_0^x 1 + x + x^2 + \frac{x^3}{2} dx$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} dx$$

Third:

$$y_3 = y_0 + \int_{x_0}^x f(x, y_2) dx$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{5} + \frac{x^5}{15} + \frac{x^6}{48}$$

$$\therefore y(0.1) = 1 + (0.1) + \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} + \frac{(0.1)^4}{15} + \frac{(0.1)^5}{48} = 1.10531$$

(20) Milne's method:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

$n \geq 03$

• predictor formula:

$$y_{n+1} = y_{n-3} + \frac{4h}{3} (2f_{n-2} - f_{n-1} + 2f_n)$$

• corrector formula:

$$y_{n+1} = y_{n-1} + \frac{h}{3} (f_{n-1} + 4f_n + f_{n+1}^p)$$

Then by Taylor's expansion, the value of y near x_0 is:

$$y(x_0 + h) = y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots$$